

Challenge (Habanero)

Q1. Solve $|x - 5| \leq 3$. Which interval notation is correct?

- (A) $[2, 8]$
 (B) $(-\infty, 2] \cup [8, \infty)$
 (C) $[5, 8]$
 (D) $[2, 5]$
 (E) $(-\infty, 2)$

Q2. In a coding competition, score s must satisfy $|s - 100| > 20$. What range of scores is allowed?

- (A) $s < 80$ or $s > 120$
 (B) $80 \leq s \leq 120$
 (C) $s \geq 100$
 (D) $s \leq 80$
 (E) $s = 100$

Q3. Graph $|x| < 4$ on a number line. What is the solution set?

- (A) $(-4, 4)$
 (B) $[-4, 4]$
 (C) $(-\infty, -4) \cup (4, \infty)$
 (D) $(-4, 0) \cup (0, 4)$
 (E) $[0, 4)$

Q4. Data usage d must satisfy $|d - 10| \geq 2$. What does this mean for the user?

- (A) $d \leq 8$ or $d \geq 12$
 (B) $8 \leq d \leq 12$
 (C) $d = 10$
 (D) $d > 10$
 (E) $d < 10$

Q5. Solve $|3x - 2| < 5$. Which inequality is equivalent?

- (A) $-1 < x < 1$
 (B) $-1 \leq x \leq 1$
 (C) $-\frac{3}{3} < x < \frac{7}{3}$
 (D) $-\frac{1}{3} < x < \frac{7}{3}$
 (E) $\frac{2}{3} < x < \frac{7}{3}$

Q6. A gamer's score s in a game must satisfy $|s| > 15$. What does this mean for the gamer?

- (A) $s < -15$ or $s > 15$
 (B) $-15 \leq s \leq 15$
 (C) $s = 0$
 (D) $s > 0$
 (E) $s < 0$

Q7. The time t it takes to complete a level must satisfy $|t - 10| \leq 3$. What range of times is allowed?

- (A) $7 \leq t \leq 13$
 (B) $t \geq 10$
 (C) $t \leq 7$
 (D) $t > 13$
 (E) $t = 10$

Q8. In a tech company, data storage d must satisfy $|d - 50| < 10$. What range of data storage is allowed?

- (A) $40 < d < 60$
 (B) $40 \leq d \leq 60$
 (C) $d \geq 60$
 (D) $d \leq 40$
 (E) $d = 50$

Open Ended Questions (FRQ)

Q9. Model the situation where a video game player's score s must satisfy $|s - 200| > 30$. If s can be any real number, what range of scores is allowed? Provide your answer in interval notation.**Q10.** Solve $|2x + 5| \leq 11$ and express the solution set in interval notation. Then, graph the solution set on a number line.

Chef's Tips

- Tip 1: Emphasize the importance of understanding absolute value inequalities and their applications.
- Tip 2: Provide examples of real-world problems that can be modeled using absolute value inequalities.
- Tip 3: Encourage students to use graphing to visualize solution sets.